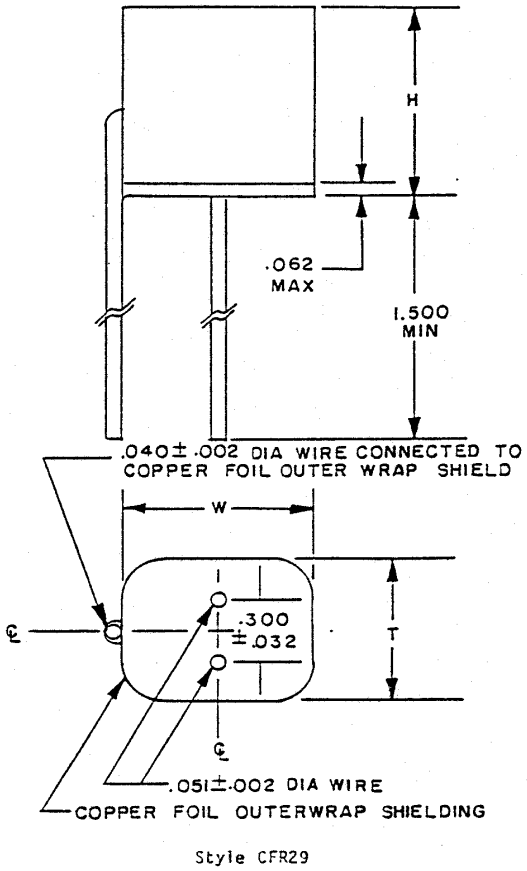
INCH-POUND
MIL-PRF-55514/12B(CR)
9 October 1997
SUPERSEDING
MIL-C-55514/12A(ER)
14 March 1990

PERFORMANCE SPECIFICATION SHEET

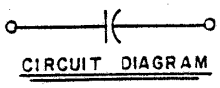
CAPACITORS, FIXED, METALLIZED PLASTIC DIELECTRIC, DC, IN NONMETAL CASES, ESTABLISHED RELIABILITY, STYLE CFR29

The specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-55514.



Style CFR29



Inches	mm
.002	0.05
.040	1.02
.046	1.17
.051	1.30
.062	1.57
.093	2.36
.300	7.62
.437	11.10
.468	11.89
.500	12.70
.562	14.27
.593	15.06
.718	18.24
.750	19.05
.781	19.84
.862	21.89
.875	22.23
.906	23.01
.937	23.80
.968	24.59
1.000	25.40
1.031	26.19
1.093	27.76
1.250	31.76
1.281	32.54
1.312	33.32
1.375	34.93
1.437	36.50
1.500	38.10
1.687	42.85
1.812	46.02

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.

FIGURE 1. Configuration and dimensions.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case: Non-hermetic.

Dielectric: Metallized polycarbonate.

Characteristic: R.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +85°C, when properly voltage derated (see figure 2), characteristic "R" capacitors will operate to +125°C.

Voltage rating: See table I.

Preconditioning: Not applicable.

Burn-in: In accordance with MIL-PRF-55514 except for the following conditions:

a. DC burn-in voltage shall be 140 percent of the dc rated voltage applied for 48 hours minimum at 125°C, +4°C, -0°C.

b. AC burn-in: Not applicable.

Insulation resistance: Insulation resistance shall be measured at rated voltage at the applicable temperature. Charge time 2 minutes maximum; however, for capacitance values greater than 1.0 µF, an additional 1 minute per µF is permitted. The following limits shall apply:

Temperature	Insulation resistance
25°C	50,000
85°C	5,000
125°C	500

Marking: In accordance with MIL-PRF-55514.

Qualification submittal:

Type designation	Sample size
CFR29RRA106FM	61
CFR29RRA506FM	61
CFR29RRG305FM	61
CFR29RRG506FM	61
CFR29RRB305FM	61
CFR29RRB506FM	61

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TABLE I. Capacitor characteristics and dimensions.

Type designation 1/	Capacitance	Capacitance tolerance	Rated voltage at 85°C	ESR milliohms at 100 Khz	Ripple current max at 25°C	Resonant frequency kHz	Dimensions		
							T 2/	W 2/	H 3/
CFR29RRA106-- CFR29RRA206-- CFR29RRA506--	10.0 µF 20.0 µF 50.0 µF	J, K, M J, K, M J, K, M	50 V 50 V 50 V	11 10 6	12.6 14.8 23.5	318 214 130	.562 .718 1.000	.862 .968 1.375	.750 .875 1.000
CFR29RRG305-- CFR29RRG505-- CFR29RRG106-- CFR29RRG206-- CFR29RRG506--	3.0 µF 5.0 µF 10.0 µF 20.0 µF 50.0 µF	J, K, M J, K, M J, K, M J, K, M J, K, M	75 V 75 V 75 V 75 V 75 V	15 13 12 11 6	10.3 11.6 13.5 16.5 24.2	672 427 301 205 120	.468 .500 .718 .937 1.312	.718 .750 1.031 1.312 1.687	.750 .875 .875 1.000 1.250
CFR29RRB305-- CFR29RRB505-- CFR29RRB106-- CFR29RRB206-- CFR29RRB506--	3.0 µF 5.0 µF 10.0 µF 20.0 µF 50.0 µF	J, K, M J, K, M J, K, M J, K, M J, K, M	100 V 100 V 100 V 100 V 100 V	15 13 12 11 6	10.6 12.4 14.6 17.1 23.6	684 427 290 192 115	.437 .593 .781 .906 1.437	.750 .906 1.093 1.281 1.812	.875 .875 1.000 1.250 1.375

1/ The complete type designation will include additional symbols to indicate capacitance tolerance and product level (C, M, P, R, and S).
2/ Tolerances: Thickness and width +.062 inch, -.156 inch (+1.57 mm, -3.96 mm) on .437 inch through .500 inch (11.10 mm through 12.7 mm) and +.093 inch, -.187 inch (+2.36 mm, -4.75 mm) on .562 inch (14.27 mm) and greater.
3/ Tolerances: Length +.093 inch, -.062 inch (+2.36 mm, -1.57 mm).

Substitutability for capacitance tolerance shall be as follows:

Capacitance tolerance	Will replace capacitance tolerance
F	G, J, K
G	J, K
J	K
K	

Substitutability for voltage shall be as follows:

Volts dc	Will replace voltages
100	75, 50
75	50

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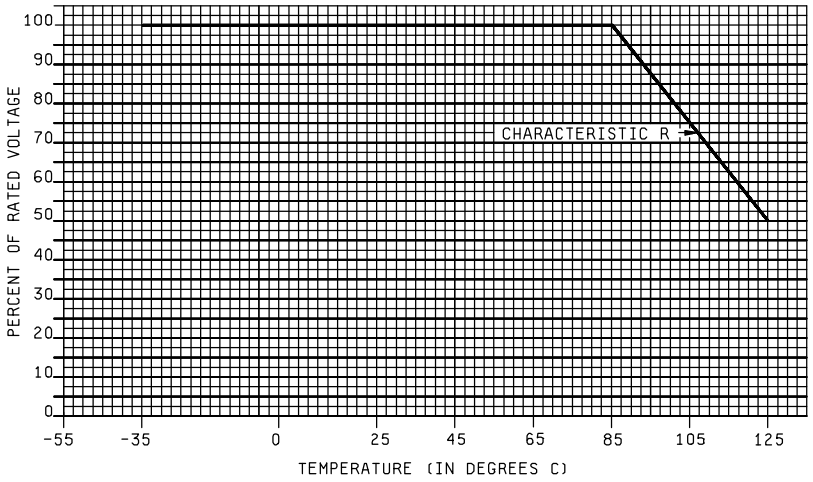

FIGURE 2. Voltage derating with temperature.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review Activities:
Navy - AS, MC, OS
Air Force - 17, 19, 99
DLA - CC

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5910-1961)

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